

Calculus

HW 5

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Sec 1



Homework

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นายภิตติพงษ์

ทัศนพรหมสิน

Find the antiderivative of the function below.

$$1) \quad f'(x) = 8x^3 + 5\sqrt[3]{x^2} - \frac{10}{x^6} + 7$$

$$f(x) = \int 8x^3 + 5\sqrt[3]{x^2} - \frac{10}{x^6} + 7 \, dx$$

$$= \int 8x^3 \, dx + \int 5x^{\frac{2}{3}} \, dx - \int 10x^{-6} \, dx + \int 7 \, dx$$

$$= \frac{8x^4}{4} + \frac{5x^{\frac{5}{3}}}{\frac{5}{3}} - \frac{10x^{-5}}{-5} + 7x$$

$$= 2x^4 + 3x\sqrt[3]{x^2} + \frac{2}{x^5} + 7x + C$$



$$2) \quad g'(x) = \frac{3 \cos x}{\sin^2 x}$$

$$g(x) = \int \frac{3 \cos x}{\sin^2 x} dx$$

$$= 3 \int \frac{1}{\sin(x)} \cdot \frac{\cos(x)}{\sin(x)} dx$$

$$= 3 \int \csc(x) \cdot \cot(x) dx$$

$$= -3 \csc(x) + C$$



$$3) \quad y' = \frac{x^2 + 3\sqrt{x}}{x^3}$$

$$y = \int \frac{x^2 + 3\sqrt{x}}{x^3} dx$$

$$= \int \frac{x^2}{x^3} + \frac{3x^{\frac{1}{2}}}{x^3} dx$$

$$= \int \frac{1}{x} + 3x^{-\frac{5}{2}} dx$$

$$= \ln|x| + \left(\frac{3x^{-\frac{3}{2}}}{-\frac{3}{2}} \right) + C$$

$$= \ln|x| - \frac{2}{x\sqrt{x}} + C$$

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